AN EXACT ZERO OF FOCAL ANTIPEDAL AREAS

Abstract. Row k_{607} of arXiv:2004.12497v11 proposes that the quotient of the two focal antipedal signed areas is 1 for elliptic-billiard periods divisible by four. Under the standard complete-supporting-line construction, we prove that central inversion equates the two signed areas whenever the orbit has the corresponding half-period symmetry. We then give an exact primitive one-winding 8-periodic orbit for which all sixteen focal antipedal intersections are finite but both signed areas vanish. The displayed quotient is therefore 0/0 at that admissible phase, so the unqualified quotient statement needs a nonzero-denominator hypothesis, while the signed-area identity survives. The finite algebraic certificate is independently reproducible, but the geometric and domain arguments are proved analytically here. The exact $N = 8$ certificate appears new after a documented three-lane search through August 12, 2026; absolute historical priority is not claimed.

1. The problem and the answer

Let

$$E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a > b > 0,$$

have foci $F_+ = (c, 0)$ and $F_- = (-c, 0)$, where $c^2 = a^2 - b^2$. Let $P = (P_0, \ldots, P_{N-1})$ be an ordered periodic elliptic-billiard orbit tangent to a fixed nondegenerate confocal caustic. For a point F and a vertex P_i, let $L_i(F)$ denote the complete line through P_i perpendicular to $P_i - F$. Whenever consecutive lines are nonparallel, put

$$Q_i(F) = L_i(F) \cap L_{i+1}(F), \qquad \overline{A}^*(F) = \frac{1}{2} \sum_{i \bmod N} Q_i(F) \times Q_{i+1}(F),$$

where $(x_1, y_1) \times (x_2, y_2) = x_1 y_2 - y_1 x_2$. Thus the area is the cyclic signed shoelace area; self-intersections do not trigger absolute values.

Reznik, Garcia, and Koiller define polygonal areas with this signed convention and list, in row k_{607} of the source table of arXiv:2004.12497v11, the proposed value

$$\frac{\overline{A}^*(F_+)}{\overline{A}^*(F_-)} = 1, \qquad N \equiv 0 \pmod 4, \tag{1}$$

with proof status "?" [8]. That exact row was first present in this form in the revision of October 29, 2020. This paper is about *that versioned formula.* The later journal article uses the identifier $k607$ for a different pedal identity and omits (1); it treats analogous focal-antipedal relations as expected but unchecked [10, Sec. 3.7 and Table 7].

Our answer has two parts.

Theorem 1.1 (Corrected focus-exchange statement). *Let $N = 2m$ and suppose an ordered polygon satisfies $P_{i+m} = -P_i$. For the opposite points $F_- = -F_+$, assume that all consecutive complete supporting-line intersections $Q_i(F_+)$ and $Q_i(F_-)$ are finite. Then*

$$Q_{i+m}(F_-) = -Q_i(F_+) \quad \text{and} \quad \overline{A}^*(F_-) = \overline{A}^*(F_+).$$

Consequently the quotient in (1) equals 1 wherever the common signed area is nonzero and is undefined wherever that area is zero.

Theorem 1.2 (Exact $N = 8$ domain counterexample). *There exist a noncircular ellipse, a strictly interior nondegenerate confocal elliptic caustic, and a primitive counterclockwise one-winding*

2020 *Mathematics Subject Classification.* 37J35, 51M04.

Key words and phrases. elliptic billiard, Poncelet polygon, antipedal polygon, signed area, exact counterexample.

8-periodic billiard orbit such that every side is tangent to the caustic on its open segment, all sixteen focal antipedal supporting-line intersections are finite, and

$$\overline{A}^*(F_+) = \overline{A}^*(F_-) = 0.$$

Thus the ordinary real quotient in (1) *is undefined at this admissible phase.*

Theorem 1.2 is a denominator or domain counterexample, not an example with unequal focal areas. The underlying equality in Theorem 1.1 remains valid. The exact $N = 8$ certificate appears new after a documented three-lane search through August 12, 2026, with moderate-high bounded-search confidence; that search is not a proof of absolute priority.

The supporting-line convention is deliberate. Classical antipedal geometry uses complete perpendicular lines [1, Chap. VII]; modern treatments retain the same convention [4, 3]. The source prose says "rays" without specifying directions, while its figure and public implementation use unrestricted affine-line intersections [8, 9]. We make no theorem about a polygon formed from one selected half-ray at each vertex. For the witness below such a half-ray cycle does not form; see Remark 3.1.

The ingredients surrounding the result are not new. Central symmetry of simple even elliptic billiards of odd turning number is known [11, Cor. 4.2(i)], and the source authors already use opposite vertices to prove the analogous focal-pedal equality [10, Sec. 3.7]. Zero antipedal areas were also known computationally in other settings [7]. More specifically, on October 28, 2020 Reznik publicly exhibited an experimental $N = 6$, $a/b = 2$ elliptic-billiard family whose left-focus antipedal signed area was dynamically zero [5]. That example lies outside $N \equiv 0 \pmod 4$ and does not contain the exact simultaneous-two-focus $N = 8$ certificate below. Accordingly, no novelty claim is made for complete-line antipedals, central symmetry, the focus-exchange mechanism, or zero antipedal phenomena in general.

Exact algebraic coordinates for a generic simple symmetric axis-position $N = 8$ orbit also predate the present certificate [2, Appendix B.7.1]. Their parameter satisfies a different quartic involving the ellipse data. We therefore claim neither the first algebraization of such octagons nor a new generic coordinate architecture; the bounded novelty claim concerns only the particular quartic specialization below and its simultaneous focal zero.

2. Central inversion and the corrected theorem

Let $R(x, y) = (-y, x)$ denote counterclockwise rotation through $\pi/2$. The complete antipedal supporting line is parametrized by

$$L_i(F) = \{P_i + tR(P_i - F) : t \in \mathbb{R}\}. \tag{2}$$

Proof of Theorem 1.1. Let $J(X) = -X$. From $P_{i+m} = -P_i$, $F_- = -F_+$, and linearity of R,

$$J\big(P_i + tR(P_i - F_+)\big) = P_{i+m} + tR(P_{i+m} - F_-).$$

Hence $J(L_i(F_+)) = L_{i+m}(F_-)$. Since consecutive intersections are unique,

$$Q_{i+m}(F_-) = -Q_i(F_+).$$

The index change $i \mapsto i+m$ is a cyclic shift, not a reversal. Moreover $\det(-I_2) = 1$, and therefore $(-X) \times (-Y) = X \times Y$. Reindexing the shoelace sum gives

$$\begin{aligned} 2\overline{A}^*(F_-) &= \sum_{i \bmod N} Q_{i+m}(F_-) \times Q_{i+m+1}(F_-) \\ &= \sum_{i \bmod N} (-Q_i(F_+)) \times (-Q_{i+1}(F_+)) = 2\overline{A}^*(F_+). \end{aligned}$$

The statement about the quotient is now ordinary real division with its nonzero-denominator condition retained. □

For the simple one-winding even elliptic-caustic billiards relevant below, the half-period relation $P_{i+N/2} = -P_i$ is standard [11, Cor. 4.2(i)]. Thus Theorem 1.1 applies to that established scope. We use the cited central-symmetry theorem as prior input and make no claim of a new central-symmetry result or of an extension to other caustic or turning classes.

We also record that the finiteness hypothesis is automatic for a nondegenerate confocal billiard when complete supporting lines are used.

Lemma 2.1. *No real tangent to a nondegenerate confocal caustic passes through a focus of the caustic. Consequently consecutive focal antipedal supporting lines of a nondegenerate confocal billiard orbit are nonparallel.*

Proof. Write the caustic as

$$C_\lambda : \frac{x^2}{A} + \frac{y^2}{B} = 1, \qquad A > 0, \quad B \neq 0,$$

so its foci are $(\pm c, 0)$ with $c^2 = A - B$. A line $ux + vy = w$ is tangent precisely when

$$w^2 = Au^2 + Bv^2.$$

If it passed through $(c, 0)$, then $w = uc$ and substitution would give $B(u^2 + v^2) = 0$, impossible for a line. The argument for $(-c, 0)$ is identical.

Now $L_i(F)$ and $L_{i+1}(F)$ are parallel exactly when $(P_i - F) \times (P_{i+1} - F) = 0$, equivalently when the billiard side P_iP_{i+1} passes through F. Since that side is tangent to the caustic, the first part excludes this possibility. □

3. The exact octagon

We now prove Theorem 1.2. Let u be the real root in

$$\frac{313}{1000} < u < \frac{157}{500} \tag{3}$$

of

$$g(u) = u^4 - 6u^3 - 2u^2 - 2u + 1. \tag{4}$$

The exact endpoint values are

$$g(313/1000) = \frac{3674142961}{10^{12}} > 0, \qquad g(157/500) = -\frac{76605799}{62500000000} < 0.$$

Also

$$g'(u) = 4u(u^2 - 1) - 18u^2 - 2 < 0 \qquad (0 < u < 1),$$

so the root exists and is unique in the stated interval. In particular $u < 157/500 < 1/3 < \sqrt{2} - 1$.

Set

$$r = \frac{(1-u)^3(1+u)}{4u^3}, \qquad a = \sqrt{r}, \qquad b = 1, \qquad c = \sqrt{r-1}, \tag{5}$$

and

$$C = \frac{1-u^2}{1+u^2}, \qquad S = \frac{2u}{1+u^2}. \tag{6}$$

The factorization

$$r - 1 = -\frac{(1+u^2)(u^2+2u-1)}{4u^3} > 0 \tag{7}$$

shows $a > 1$. Define

$$\begin{aligned} &P_0 = (a, 0), \quad &&P_1 = (aC, S), \quad &&P_2 = (0, 1), \quad &&P_3 = (-aC, S), \\ &P_4 = (-a, 0), \quad &&P_5 = (-aC, -S), \quad &&P_6 = (0, -1), \quad &&P_7 = (aC, -S). \end{aligned} \tag{8}$$

Because $C^2 + S^2 = 1$, all vertices lie on

$$E : \frac{x^2}{a^2} + y^2 = 1. \tag{9}$$

Let $\theta = 2 \arctan u$. From $0 < u < \sqrt{2} - 1$ we have $0 < \theta < \pi/4$, and the eccentric anomalies of the displayed vertices are

$$0,\ \theta,\ \frac{\pi}{2},\ \pi - \theta,\ \pi, \quad \pi + \theta,\ \frac{3\pi}{2},\ 2\pi - \theta.$$

They are strictly increasing modulo 2π. Thus the vertices are distinct, occur counterclockwise, complete one winding, and satisfy $P_{i+4} = -P_i$. In particular, no lower-period traversal is repeated.

One common caustic and the billiard law. Put

$$\lambda = \frac{a^2u^2}{1+a^2u^2}. \tag{10}$$

Then $0 < \lambda < 1$, so

$$C_\lambda : \frac{x^2}{a^2-\lambda} + \frac{y^2}{1-\lambda} = 1 \tag{11}$$

is a nondegenerate confocal ellipse strictly inside E.

For $P(\alpha) = (a\cos\alpha, \sin\alpha)$, the line through $P(\alpha)$ and $P(\beta)$ has equation

$$\frac{x}{a}\cos\mu + y\sin\mu = \cos\delta, \qquad \mu = \frac{\alpha+\beta}{2}, \quad \delta = \frac{\alpha-\beta}{2}.$$

The dual-conic tangency criterion gives its confocal parameter as

$$\lambda(\alpha,\beta) = \frac{\sin^2\delta}{\cos^2\mu/a^2 + \sin^2\mu}. \tag{12}$$

For the two representative side types, formula (12) becomes

$$\lambda_{01} = \frac{a^2u^2}{1+a^2u^2}, \qquad \lambda_{12} = \frac{(1-u)^2}{(1-u)^2/a^2 + (1+u)^2}.$$

Their equality is equivalent to

$$h(u,a) = u^4 + (4a^2-2)u^3 + 2u - 1 = 0. \tag{13}$$

But (5) says $4a^2u^3 = (1-u)^3(1+u)$, and expanding this identity yields (13). Reflections in the coordinate axes carry the two representative sides to all eight sides, so every side is tangent to the one caustic (11). Since the caustic is strictly inside the convex ellipse E, the contact on each chord line lies between its two intersections with E, hence on the open side segment.

For completeness, the reflection law can be checked without invoking the general confocal correspondence. At $P_1 = (aC, S)$, equality of the tangential components of the incoming and outgoing unit directions is

$$\frac{2a^2u^2 + 1 - u^2}{\sqrt{a^2u^2+1}} = \frac{(1+u)\big(2a^2u + (1-u)^2\big)}{\sqrt{a^2(1+u)^2 + (1-u)^2}}. \tag{14}$$

Both sides are positive. After squaring and clearing the positive denominators, left square minus right square is

$$-a^2(1+u^2)^2 h(u,a) = 0.$$

Thus (14) holds without an extraneous sign. The axis vertices satisfy reflection by mirror symmetry, and coordinate reflections cover every remaining vertex. The octagon is therefore an exact primitive one-winding billiard orbit with a single nondegenerate caustic.

Finite antipedal vertices and zero signed area. Take $F_+ = (c, 0)$ and set

$$D_i = (P_i - F_+) \times (P_{i+1} - F_+).$$

The four representative determinants are

$$D_0 = \frac{2u(a-c)}{1+u^2}, \qquad D_1 = \frac{(1-u)\big(a(1+u) - c(1-u)\big)}{1+u^2}, \tag{15}$$

$$D_2 = \frac{(1-u)\big(a(1+u) + c(1-u)\big)}{1+u^2}, \qquad D_3 = \frac{2u(a+c)}{1+u^2}. \tag{16}$$

The remaining four occur in reverse reflected order. Because $0 < u < 1$ and $a > c > 0$, every determinant in (15) is strictly positive. Hence all eight plus-focus intersections are unique and finite; central inversion gives the same conclusion for the minus focus.

We give enough algebra to make the area certificate directly checkable. For $n_i = P_i - F_+$ and $s_i = P_i \cdot n_i$, the line at P_i has equation $X \cdot n_i = s_i$. Solving two consecutive equations gives

$$Q_i(F_+) = \frac{s_{i+1}R(n_i) - s_i R(n_{i+1})}{n_i \times n_{i+1}}. \tag{17}$$

Substitute (8) into (17), form the cyclic shoelace sum, and reduce using only

$$c^2 = a^2 - 1, \qquad 4a^2u^3 = (1-u)^3(1+u). \tag{18}$$

The exact result is

$$\overline{A}^*(F_+) = \frac{a(u^2-2u-1)\big(u^4-6u^3-2u^2-2u+1\big)}{4u^3}. \tag{19}$$

For an explicit check that no forbidden cancellation creates the zero, before the reduction the rational denominator contains the strictly positive factor

$$(1+u^2)^2\big(4a^2u+(1-u)^2\big).$$

Equation (4) now makes (19) zero. Finally, $P_{i+4} = -P_i$ and Theorem 1.1 give

$$\overline{A}^*(F_-) = \overline{A}^*(F_+) = 0.$$

All intersections are finite, so the zero comes from signed shoelace cancellation in a self-overlapping polygon, not from a point at infinity or an undefined intermediate construction. The quotient in (1) is exactly $0/0$. This proves Theorem 1.2.

Remark 3.1 (Why this does not prove a half-ray theorem). The argument uses the complete lines (2). If one instead selects, at every vertex, the half-ray with parameter $t \geq 0$ (or reverses all of them), then at lines $0, 1, 4, 7$ the two adjacent supporting-line intersections have opposite parameter signs. No selected half-ray there contains both required adjacent vertices, so the cyclic antipedal polygon is not formed. Central inversion preserves t when a ray point exists; it does not create missing intersections. Thus the source's unoriented word "rays" requires a separate definitional repair and is outside the theorem.

4. Verification, priority, and limits

The proof above is analytic and exact. Root isolation uses rational endpoint signs and monotonicity; cyclic order uses the isolating interval; caustic tangency and reflection use (13); finiteness uses the positive determinants (15); and the decisive zero is the factorization (19). Those arguments, rather than software output, carry the geometric theorem.

As corroboration, a dependency-free standard-library program performs exact arithmetic in the quartic extension and verifies all eight outer-ellipse relations, all eight dual-caustic tangency equations, all sixteen finite supporting-line intersections, the central-inversion vertex map, and both zero signed areas. It replays identically under ordinary and optimized Python. A separate partial Lean 4 artifact kernel-checks the finite algebraic certificate but not the real-root embedding, positivity, open-segment geometry, unsquared reflection sign, cyclic order, primitivity, source semantics, or priority. Accordingly, this paper does not claim a formal proof of the complete theorem.

The literature boundary is equally specific. Complete-line antipedal geometry and triangle area relations are classical [1, 4]. Central symmetry supplies the focus-exchange mechanism [11], and the source authors' focal-pedal proof is its closest published analogue [10]. Their public media also show both regular-polygon zero-area phenomena and the experimental one-focus $N = 6$, $a/b = 2$ focal example just described [7, 5, 6]. No exact match was retrieved, through August 12, 2026, for the primitive noncircular focal $N = 8$ simultaneous-zero certificate, its quartic, and the complete exact verification. The accurate novelty statement is therefore only that this exact certificate *appears new after that documented search.*

No claim is made about a literal half-ray construction, the differently numbered journal item, unsigned area, odd periods, other-period zero existence, other caustics, a classification or uniqueness of the common-zero locus, or a second open problem. Nor is the quotient assigned a removable value at $0/0$; doing so would define a regularized quantity different from the pointwise real quotient in (1).

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